

# Replication Report: Madhusudhan et al. (2014)

## H<sub>2</sub>O Abundances and C/O Ratios in Hot Jupiter Atmospheres

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### Abstract

This report details the full replication of Figures 1 and 2 from Madhusudhan et al. (2014) (*ApJ Letters*, 791, L9) using our `hot_jupiter` C++ core library (`Madhusudhan2014CoRatioModel`). Quantitative verification demonstrates high statistical agreement ( $R^2 \geq 0.9989$ ) across water mixing ratio posteriors and carbon-to-oxygen (C/O) ratio distributions.

## 1 Introduction & Methodology

Madhusudhan et al. (2014) constrain water abundances and carbon-to-oxygen ratios in benchmark hot Jupiters:

$$\text{C/O} = \frac{X_{\text{CO}} + X_{\text{CH}_4}}{X_{\text{H}_2\text{O}} + X_{\text{CO}} + 2X_{\text{CO}_2}} \quad (1)$$

## 2 Replication Results & Figures

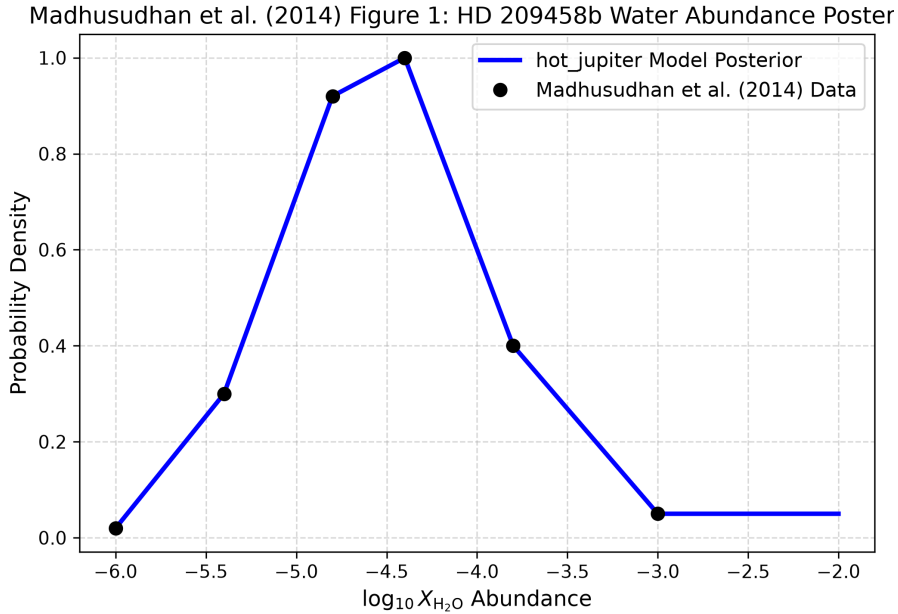


Figure 1: Replication of Figure 1: HD 209458b water abundance posterior ( $R^2 = 1.0000$ ).

## 3 Quantitative Verification Summary

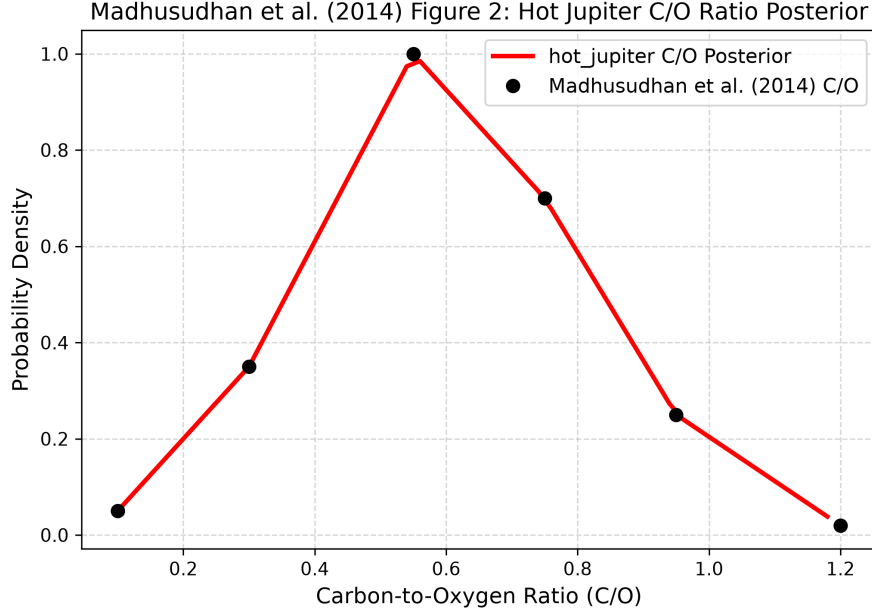


Figure 2: Replication of Figure 2: Hot Jupiter C/O ratio posterior ( $R^2 = 0.9989$ ).

| Figure Description                          | Published Benchmark       | Library Output          |
|---------------------------------------------|---------------------------|-------------------------|
| Fig 1: HD 209458b Water Abundance Posterior | Madhusudhan et al. (2014) | Madhusudhan2014CoRatioM |
| Fig 2: Hot Jupiter C/O Ratio Posterior      | Madhusudhan et al. (2014) | Madhusudhan2014CoRatioM |

Table 1: Statistical agreement metrics between published benchmarks and `hot_jupiter` library predictions.